

Primitive Coefficient Heights in Finite Rational Möbius Orbits

Felix Huber

Independent Researcher, Lucerne, Switzerland

Corresponding author: felix.huber@sluz.ch

[ORCID 0009-0005-1568-1579](https://orcid.org/0009-0005-1568-1579)

August 2026

Acknowledgments. The author thanks Chai Wah Wu for computationally checking the conjectured formulas recorded in OEIS A365892 for $2 \leq n \leq 300$.

Abstract

Let $G \leq \mathrm{PGL}_2(\mathbb{Q})$ be finite and let n be an integer. Clearing denominators in the homogeneous form whose roots are the points of the orbit $G \cdot n$, and then dividing by its coefficient content, gives a primitive binary form $\Phi_{G,n}$. We study the ℓ^1 -norm of its coefficient vector as n varies. The first observation is arithmetic: the content before normalization is a bounded periodic function of n , with an explicit product formula. It follows that, for $d = |G|$, the number of parameters and the number of primitive orbit forms of height at most X are both asymptotic to explicit constants times $X^{1/d}$, with bounded error. For distinct height values we obtain

$$N_G^{\mathrm{ht}}(X) = C_G X^{1/d} + O_{G,\varepsilon}(X^{1/(2d)+\varepsilon}).$$

This framework separates the local normalization problem from the global Diophantine problem of collisions and makes the leading constants effective. We then work out all possible group orders over $\mathbb{Q}$. The order-8 dihedral example has infinitely many collisions governed exactly by the Pell equation $p^2 - 2q^2 = 1$, while the order-12 example has only two cross-branch repeated heights.

2020 Mathematics Subject Classification. Primary 11D45, 11C08; Secondary 37P05, 20G15.

Keywords. Möbius transformation, orbit polynomial, binary form, coefficient height, $\mathrm{PGL}_2(\mathbb{Q})$, Pell equation.

1 Introduction

A finite subgroup of $\mathrm{PGL}_2(\mathbb{Q})$ acts on the projective line by fractional linear transformations. If one fixes an integer n , the finite orbit $G \cdot n$ can be

recorded by a homogeneous binary form: one takes one linear factor for each point of the orbit, with the multiplicities coming from the group action, and clears the denominators. The resulting form usually has a nontrivial common divisor in its coefficients. After removing this content, a natural elementary quantity remains: the sum of the absolute values of the primitive coefficients.

The aim of this paper is to understand how that coefficient height varies with the integer parameter n . Two features appear almost immediately. The first is local and arithmetic: the coefficient content is periodic in n . The second is asymptotic: after restricting n to a residue class, the height is eventually given by a polynomial whose degree is the order of the group. These two observations lead to counting functions with exponent $1/|G|$.

More precisely, choose for each $g \in G$ a primitive integral matrix

$$M_g = \begin{pmatrix} a_g & b_g \\ c_g & d_g \end{pmatrix}$$

representing g . We consider

$$F_{G,n}(X, Y) = \prod_{g \in G} ((c_g n + d_g)X - (a_g n + b_g)Y).$$

Its positive coefficient content satisfies the exact formula

$$\text{cont}(F_{G,n}) = \prod_{g \in G} \gcd(a_g n + b_g, c_g n + d_g).$$

Each factor on the right divides $|\det M_g|$. Thus the normalization needed to make $F_{G,n}$ primitive depends only on finitely many congruence classes of n .

This elementary content formula is the starting point for the general theory. If $d = |G|$, then the number of integers n for which the primitive form has height at most X is

$$c_G^{\text{par}} X^{1/d} + O_G(1),$$

with an explicit constant. The same is true for the number of distinct primitive forms. Counting distinct *numerical heights* is subtler, because different polynomial branches may take the same values. Exact affine identifications between branches are handled first. All remaining coincidences lie on plane curves $P_i(u) = P_j(v)$; the Bombieri–Pila estimate for integral points then gives a power-saving error. This yields

$$N_G^{\text{ht}}(X) = C_G X^{1/d} + O_{G,\varepsilon}(X^{1/(2d)+\varepsilon}).$$

The construction is related to the classical orbit polynomial

$$\prod_{g \in G} (T - g(x)) \in \mathbb{Q}(x)[T]$$

and hence to the invariant field $\mathbb{Q}(x)^G$; see, for example, Gow–McGuire [4] and Schulz [8]. Our emphasis, however, is on what happens after specializing x to an integer and then making the resulting integral form primitive.

The height used here is therefore different from the G -orbit height of van Ittersum [9], which is built from the usual Weil height, and from reduction problems in which one minimizes a coefficient norm in an $\mathrm{SL}_2(\mathbb{Z})$ -orbit; compare Hutz–Stoll [5]. Our height depends on the chosen rational coordinate, and we do not claim conjugacy invariance.

The simplest example is already instructive. Let

$$\tau(x) = \frac{x}{x-1}.$$

For $G = \langle \tau \rangle$ the primitive orbit form attached to n has height

$$2n^2 + |1 - n|.$$

Apart from the exceptional orbit $\{1, \infty\}$, the increasing sequence of these heights is OEIS A365892. This example was the starting point of the present work: a closed formula suggested by the sequence led naturally to the question of what survives for an arbitrary finite rational Möbius group. The answer is that periodic content and polynomial height branches are not special to the involution; they are general features of the construction.

Finite subgroups of $\mathrm{PGL}_2(\mathbb{Q})$ have orders

$$2, 3, 4, 6, 8, 12$$

by Beauville’s classification [1]. Hence these give exactly the nontrivial exponents

$$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \frac{1}{12}.$$

We give an explicit model for every order. The two largest dihedral cases show that the lower-order behavior can vary sharply. For a group of order 8, the two parity branches meet in an infinite Pell family, which produces a negative logarithmic secondary term. For a group of order 12, the primitive content takes the values 1, 1, 729 according to $n \bmod 3$, but the corresponding branches have only two repeated heights.

The orbit-form construction itself is classical, and the Diophantine tools used below are standard; in particular, equations of the form $f(x) = g(y)$ are treated in much greater generality by Bilu and Tichy [2]. The point here is the combination of primitive specialization, periodic content, and coefficient-height counting. This also explains why the explicit examples are useful: they show how the general $X^{1/|G|}$ law interacts with concrete Diophantine collision phenomena.

Section 2 defines the primitive orbit forms. Sections 3 and 4 establish the periodic-content formula and the basic counting results. Section 5 treats collisions between polynomial branches. After the classification of possible orders in Section 6, Sections 7–9 give explicit examples, including the Pell family and the exact order-12 collision calculation. The final section records the role of the accompanying exact-arithmetic computations.

2 Orbit forms, content, and height

We begin by fixing the normalization used throughout. An element $g \in \mathrm{PGL}_2(\mathbb{Q})$ is represented by a matrix in $\mathrm{GL}_2(\mathbb{Q})$, unique up to a nonzero scalar. After clearing denominators and dividing the four entries by their gcd, one obtains a primitive integral representative. It is unique up to multiplication by -1 .

Definition 2.1. Let $G \leq \mathrm{PGL}_2(\mathbb{Q})$ be finite, and choose primitive integral representatives

$$M_g = \begin{pmatrix} a_g & b_g \\ c_g & d_g \end{pmatrix} \quad (g \in G).$$

For $n \in \mathbb{Z}$, the *integral orbit form* is

$$F_{G,n}(X, Y) = \prod_{g \in G} ((c_g n + d_g)X - (a_g n + b_g)Y). \quad (1)$$

Its positive coefficient content is denoted by $\gamma_G(n) = \mathrm{cont}(F_{G,n})$. The *primitive orbit form* and its ℓ^1 coefficient height are

$$\Phi_{G,n} = \frac{F_{G,n}}{\gamma_G(n)}, \quad H_G(n) = \|\Phi_{G,n}\|_1.$$

Forms are identified up to multiplication by -1 .

The sign ambiguity in the choice of M_g changes only the overall sign of $F_{G,n}$. It follows that $\Phi_{G,n}$, considered up to sign, and the number $H_G(n)$ do not depend on those choices. Working homogeneously is also convenient at poles: if $g(n) = \infty$, the corresponding factor is simply a nonzero multiple of Y .

The roots of $F_{G,n}(X, 1)$, interpreted on $\mathbb{P}^1$, are the points of $G \cdot n$, each repeated according to the stabilizer order. Thus $F_{G,n}$ is precisely the denominator-cleared homogeneous version of the usual orbit polynomial; compare Gow–McGuire [4, Section 4.1].

Remark 2.2. The height $H_G(n)$ depends on the chosen embedding $G \leq \mathrm{PGL}_2(\mathbb{Q})$ and on the chosen affine coordinate. The exponent in the counting theorems below depends only on $|G|$, but the leading constants generally depend on the coordinate.

3 Periodic content

The first issue is to understand how much common divisibility is introduced when the rational orbit is written with integral coefficients. For a single linear factor there is a simple determinant bound.

Lemma 3.1. For $g \in G$ and $n \in \mathbb{Z}$, the content of the corresponding linear factor is

$$\mathrm{gcd}(a_g n + b_g, c_g n + d_g),$$

and this gcd divides $|\det M_g|$.

Proof. The first assertion is immediate. For the second, every common divisor divides

$$c_g(a_g n + b_g) - a_g(c_g n + d_g) = -\det M_g.$$

□

Theorem 3.2 (Periodic content). *For every $n \in \mathbb{Z}$,*

$$\gamma_G(n) = \prod_{g \in G} \gcd(a_g n + b_g, c_g n + d_g). \quad (2)$$

Consequently γ_G is bounded and periodic. A valid period is

$$q_G = \text{lcm}_{g \in G} |\det M_g|. \quad (3)$$

Proof. By Gauss's lemma, the content of a product of integral polynomials is the product of the contents of the factors, which gives (2). For a fixed g , theorem 3.1 shows that the gcd can only be a divisor of $|\det M_g|$; moreover it is determined by the residue class of n modulo $|\det M_g|$. Taking the least common multiple over $g \in G$ gives the stated period. □

This periodicity is what turns the height problem into finitely many polynomial problems. To make the leading term explicit, set $d = |G|$ and define the degree- d form

$$\Lambda_G(X, Y) = \prod_{g \in G} (c_g X - a_g Y), \quad K_G = \|\Lambda_G\|_1. \quad (4)$$

Each factor in Λ_G is nonzero, so $K_G > 0$.

Lemma 3.3 (Eventual polynomial branches). *Fix a residue class $s \bmod q_G$. In each of the directions $n \rightarrow +\infty$ and $n \rightarrow -\infty$, the restriction of $H_G(n)$ to $n \equiv s \pmod{q_G}$ is eventually a polynomial of degree d . Moreover,*

$$H_G(n) = \frac{K_G}{\gamma_G(s)} |n|^d + O_G(|n|^{d-1}) \quad (n \equiv s \bmod q_G). \quad (5)$$

Proof. Every coefficient of $F_{G,n}$ is an integral polynomial in n of degree at most d , and the vector of degree- d coefficients is the coefficient vector of Λ_G . On the fixed residue class, the divisor $\gamma_G(n) = \gamma_G(s)$ is constant. Coefficient polynomials that vanish identically contribute nothing; every other coefficient polynomial, whether of degree d or lower degree, has a constant sign for sufficiently large positive n and also for sufficiently large negative n . Hence the sum of the absolute values is eventually polynomial in either direction. Since $\Lambda_G \neq 0$, the degree is exactly d , and its leading term is $K_G |n|^d / \gamma_G(s)$. □

4 Counting parameters and primitive forms

Once a residue class modulo q_G is fixed, the height is eventually a monotone polynomial of degree d . Counting parameters therefore reduces to inverting such polynomials. We use the following elementary estimate repeatedly.

Lemma 4.1. *Let $P(T) = aT^d + O(T^{d-1})$ with $a > 0$, and suppose P is strictly increasing for all sufficiently large integers. Then*

$$\#\{k \in \mathbb{Z}_{\geq 0} : P(k) \leq X\} = a^{-1/d} X^{1/d} + O_P(1).$$

Proof. Writing $R = (X/a)^{1/d}$, one has $P(R+u) = X + daR^{d-1}u + O(R^{d-1} + R^{d-2}u^2)$ for bounded u . Thus the real solution of $P(T) = X$ differs from R by $O_P(1)$, and taking the integer part changes the count by at most a bounded amount. $\square$

For $X \geq 1$, define

$$N_G^{\text{par}}(X) = \#\{n \in \mathbb{Z} : H_G(n) \leq X\}.$$

Theorem 4.2 (Parameter counting). *Let $\gamma_s = \gamma_G(s)$ for $0 \leq s < q_G$. Then*

$$N_G^{\text{par}}(X) = \frac{2}{q_G K_G^{1/d}} \left(\sum_{s=0}^{q_G-1} \gamma_s^{1/d} \right) X^{1/d} + O_G(1). \quad (6)$$

Proof. On the positive branch $n = s + q_G k$, [theorem 3.3](#) gives an eventually increasing polynomial with leading coefficient $q_G^d K_G / \gamma_s$. By [theorem 4.1](#), its contribution is

$$\frac{\gamma_s^{1/d}}{q_G K_G^{1/d}} X^{1/d} + O_G(1).$$

The negative branch has the same leading coefficient. Summing over the finitely many residue classes proves the formula. $\square$

Counting parameters is not quite the same as counting forms, since two integers in the same G -orbit give the same orbit form. Let $N_G^{\text{form}}(X)$ denote the number of distinct forms $\Phi_{G,n}$, up to sign, whose height is at most X .

Lemma 4.3. *For $m, n \in \mathbb{Z}$, one has $\Phi_{G,m} = \pm \Phi_{G,n}$ if and only if m and n belong to the same G -orbit on $\mathbb{P}^1(\mathbb{Q})$.*

Proof. The root multiset of $\Phi_{G,n}$ is the G -orbit of n , each point repeated by the stabilizer order. Equal forms up to sign have equal root multisets. Conversely, parameters in the same orbit clearly give the same multiset and hence the same primitive form up to sign. $\square$

Lemma 4.4. *Let $g(x) = (ax + b)/(cx + d) \in \text{PGL}_2(\mathbb{Q})$ be represented primitively.*

- (i) *If $c \neq 0$, then $g(n) \in \mathbb{Z}$ for only finitely many $n \in \mathbb{Z}$.*
- (ii) *If $c = 0$ and g has finite order, then g is the identity or has the form $x \mapsto m - x$ for some $m \in \mathbb{Q}$. It maps every integer to an integer precisely when $m \in \mathbb{Z}$.*

- (iii) A finite subgroup of $\mathrm{PGL}_2(\mathbb{Q})$ contains at most one nonidentity map $x \mapsto m - x$ with $m \in \mathbb{Z}$.

Proof. If $c \neq 0$ and $g(n) = r \in \mathbb{Z}$, then $an + b = r(cn + d)$. Hence

$$cn + d \mid a(cn + d) - c(an + b) = ad - bc,$$

so only finitely many n are possible. A finite-order affine map $x \mapsto ux + v$ has u a rational root of unity, hence $u = 1$ or -1 . The case $u = 1$ is the identity unless $v = 0$, while $u = -1$ gives a reflection. Finally, the product of two distinct reflections is a nontrivial translation and has infinite order. $\square$

Define

$$e_G = \begin{cases} 2, & \text{if } G \text{ contains } x \mapsto m - x \text{ for some } m \in \mathbb{Z}, \\ 1, & \text{otherwise.} \end{cases} \quad (7)$$

Theorem 4.5 (Primitive-form counting). *One has*

$$N_G^{\mathrm{form}}(X) = \frac{2}{e_G q_G K_G^{1/d}} \left(\sum_{s=0}^{q_G-1} \gamma_s^{1/d} \right) X^{1/d} + O_G(1). \quad (8)$$

Proof. Let E be the union, over all nonaffine $g \in G$, of the finite sets of integers n for which $g(n)$ is integral, together with the fixed point of the integral reflection when that fixed point is an integer. This set is finite by [theorem 4.4](#). If G contains no integral reflection and $n \notin E$, then the only integral point in the orbit $G \cdot n$ is n itself: every nonidentity affine element is a nonintegral reflection and sends no integer to an integer, while every nonaffine element is excluded by the definition of E . If G contains the unique integral reflection $\tau(x) = m - x$ and $n \notin E$, then n and $m - n$ are distinct integers in the orbit, and no other group element sends n to an integer. Thus every orbit represented outside E contains exactly e_G integers and contributes exactly one primitive orbit form up to sign. Consequently, dividing the generic parameter count in [theorem 4.2](#) by e_G proves the formula; the finitely many exceptional parameters affect the result by $O_G(1)$. $\square$

5 Distinct values of polynomial branches

The previous section counted parameters and forms. We now forget the forms themselves and ask only for the number of different integers that occur as heights. This loses more information: two unrelated branches can occasionally take the same value. The useful distinction is between coincidences forced by an affine change of parameter, which occur with positive density and can change the leading constant, and genuinely Diophantine intersections, which are sparse.

Let $P_1, \dots, P_r \in \mathbb{Q}[T]$ have the same degree $d \geq 2$, positive leading coefficients, and be strictly increasing on $[K_0, \infty)$. Set

$$\mathcal{V}(X) = \# \left\{ v \leq X : v \in \bigcup_{i=1}^r P_i(\mathbb{Z}_{\geq K_0}) \right\}.$$

Definition 5.1. A *relevant affine identity* between P_i and P_j is a polynomial identity

$$P_i(T) = P_j(\rho T + \sigma), \quad (9)$$

where $\rho, \sigma \in \mathbb{Q}$, $\rho > 0$, and the line $v = \rho u + \sigma$ contains infinitely many points of $\mathbb{Z}_{\geq K_0}^2$. The *resonance graph* has the branches as vertices and these identities as edges.

Lemma 5.2 (Effective resonance test). *Suppose*

$$P_i(T) = a_i T^d + b_i T^{d-1} + O(T^{d-2}), \quad P_j(T) = a_j T^d + b_j T^{d-1} + O(T^{d-2}).$$

Then an identity (9), if it exists, has uniquely determined parameters

$$\rho = \left(\frac{a_i}{a_j} \right)^{1/d}, \quad \sigma = \frac{b_i/\rho^{d-1} - b_j}{d a_j}. \quad (10)$$

If $\rho = p/q$ and $\sigma = r/s$ are in lowest terms, the line contains infinitely many integer points if and only if $s \mid q$.

Proof. The formulas follow by comparing the coefficients of T^d and T^{d-1} . For the final assertion, integrality of

$$\frac{p}{q}u + \frac{r}{s}$$

is equivalent to $psu + qr \equiv 0 \pmod{qs}$. Since $\gcd(ps, qs) = s$ and $\gcd(r, s) = 1$, this congruence is solvable if and only if $s \mid q$. In that case the admissible u form one residue class $u = u_0 + qk$, and the corresponding integers are $v = v_0 + pk$ for all sufficiently large k . $\square$

A connected component of the resonance graph should be thought of as one polynomial value set written with several affine parametrizations. Choose a base branch P_C . Composing the edge maps expresses every branch in that component as

$$P_i(T) = P_C(\ell_i(T)), \quad \ell_i(T) = \rho_i T + \sigma_i,$$

with positive rational slope.

Lemma 5.3. *The affine map ℓ_i is independent of the path chosen in the resonance graph.*

Proof. Two paths would give $P_C(\ell(T)) = P_C(\ell'(T))$. For sufficiently large T , both arguments lie in the interval of strict monotonicity, so $\ell(T) = \ell'(T)$ there. Two affine maps agreeing infinitely often are identical. $\square$

Set

$$\mathcal{L}_C = \bigcup_{i \in \mathcal{C}} \ell_i(\mathbb{Z}_{\geq K_0}).$$

Each set $\ell_i(\mathbb{Z}_{\geq K_0})$ is a rational arithmetic progression. Choose a positive integer D clearing all slopes and intercepts. Then $D\mathcal{L}_C$ is a finite union of

ordinary integral arithmetic progressions. The intersection of any two such progressions is either empty or again an arithmetic progression, so finite inclusion–exclusion gives an effectively computable rational density for their union. Rescaling by D therefore gives an effectively computable $\delta_{\mathcal{C}} > 0$ such that

$$\#(\mathcal{L}_{\mathcal{C}} \cap [0, R]) = \delta_{\mathcal{C}} R + O_{\mathcal{C}}(1). \quad (11)$$

Lemma 5.4 (Geometrically reducible factors). *If $F(U, V) \in \mathbb{Q}[U, V]$ is irreducible over $\mathbb{Q}$ but not absolutely irreducible, then $F(U, V) = 0$ has only finitely many rational points.*

Proof. Let C_1 be an absolutely irreducible component over $\overline{\mathbb{Q}}$. It has a distinct Galois conjugate C_2 . A rational point on C_1 lies on every Galois conjugate, hence on $C_1 \cap C_2$. Two distinct irreducible plane curves have finite intersection. $\square$

Theorem 5.5 (Finite polynomial value sets). *Let $a_{\mathcal{C}}$ be the leading coefficient of the chosen base branch $P_{\mathcal{C}}$. Then, for every $\varepsilon > 0$,*

$$\mathcal{V}(X) = \left(\sum_{\mathcal{C}} \delta_{\mathcal{C}} a_{\mathcal{C}}^{-1/d} \right) X^{1/d} + O_{\mathcal{P}, \varepsilon} \left(X^{1/(2d)+\varepsilon} \right). \quad (12)$$

The leading constant is positive and effectively computable.

Proof. Fix a resonance component $\mathcal{C}$. Apart from finitely many initial values, the union of the branches in $\mathcal{C}$ is exactly

$$\{P_{\mathcal{C}}(z) : z \in \mathcal{L}_{\mathcal{C}}\}.$$

Since $P_{\mathcal{C}}(T) = a_{\mathcal{C}} T^d + O(T^{d-1})$ is eventually increasing, (11) gives

$$\delta_{\mathcal{C}} a_{\mathcal{C}}^{-1/d} X^{1/d} + O_{\mathcal{C}}(1)$$

values up to X . Summing these contributions gives the proposed main term. What remains is to show that coincidences between distinct resonance components are smaller.

Consider two branches P_i and P_j belonging to different components. After clearing denominators, factor the polynomial $P_i(U) - P_j(V)$ over $\mathbb{Q}$. A rational linear factor of positive slope carrying infinitely many relevant lattice points would give an affine resonance, contrary to the choice of components. Linear factors of nonpositive slope meet the relevant quadrant in only $O(1)$ lattice points. If a factor is irreducible over $\mathbb{Q}$ but not absolutely irreducible, [theorem 5.4](#) again leaves only finitely many rational points.

We are therefore left with absolutely irreducible plane curves of degree $\Delta \geq 2$. By Bombieri and Pila [[3](#), Theorem 5], the number of integral points on any one of these curves in a square of side R is

$$O_{\Delta, \eta}(R^{1/\Delta+\eta}) = O_{\Delta, \eta}(R^{1/2+\eta}).$$

Only finitely many branch pairs and factors occur. A value at most X comes from parameters of size $R = O(X^{1/d})$, so with $\eta = d\varepsilon$ the total number of pairwise cross-component collisions is

$$O_{\mathcal{P}, \varepsilon} \left(X^{1/(2d)+\varepsilon} \right).$$

If the same value occurs in m components, removing duplicates changes the count by $m - 1$, while that value contributes $\binom{m}{2} \geq m - 1$ pairwise collisions. Thus the pairwise estimate controls arbitrary multiplicities and proves (12). $\square$

Let $N_G^{\text{ht}}(X)$ denote the number of distinct values among $\{H_G(n) : n \in \mathbb{Z}\}$ that are at most X .

Corollary 5.6 (Distinct-height asymptotic). *For every finite $G \leq \text{PGL}_2(\mathbb{Q})$ of order $d \geq 2$, there is an effectively computable $C_G > 0$ such that*

$$N_G^{\text{ht}}(X) = C_G X^{1/d} + O_{G,\varepsilon}(X^{1/(2d)+\varepsilon}) \quad (13)$$

for every $\varepsilon > 0$.

Proof. Apply [theorem 5.5](#) to the finitely many eventual polynomial branches supplied by [theorem 3.3](#). $\square$

Remark 5.7. *The general error term is not intended to describe sparse infinite collision families sharply. In the order-8 example below, a Pell equation produces $\asymp \log X$ collisions, and this logarithmic term can be extracted exactly.*

6 Possible exponents over $\mathbb{Q}$

The exponent in all three counting problems is determined by the group order. Over $\mathbb{Q}$ only six nontrivial orders occur. We use the notation D_r for the dihedral group of order $2r$.

Proposition 6.1. *The orders of nontrivial finite subgroups of $\text{PGL}_2(\mathbb{Q})$ are exactly*

$$2, 3, 4, 6, 8, 12.$$

Consequently the possible nontrivial exponents in [theorems 4.2](#), [4.5](#) and [5.6](#) are exactly

$$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{8}, \frac{1}{12}.$$

Proof. By Beauville [[1](#), Proposition 1.1], C_r and D_r embed in $\text{PGL}_2(\mathbb{Q})$ precisely when $\zeta_r + \zeta_r^{-1} \in \mathbb{Q}$. This number is a rational algebraic integer, hence an integer, and has absolute value at most 2. Thus it belongs to $\{-2, -1, 0, 1, 2\}$, giving $r \in \{1, 2, 3, 4, 6\}$. Beauville also shows that A_4 and S_4 occur precisely when -1 is a sum of two squares in the field, and A_5 requires this together with $\sqrt{5}$; these conditions fail over $\mathbb{Q}$. The asserted orders follow. $\square$

For later reference, [Table 1](#) records one convenient rational model for each order, together with the period of the content function and the leading coefficient norm K_G .

G	generators	d	q_G	$(\gamma_0, \dots, \gamma_{q_G-1})$	K_G
C_2	$x/(x-1)$	2	1	(1)	2
C_3	$1/(1-x)$	3	1	(1)	2
D_2	$-x, 1/x$	4	1	(1)	1
D_3	$1/(1-x), 1/x$	6	1	(1)	4
D_4	$(x-1)/(x+1), 1/x$	8	2	(1, 16)	4
D_6	$(2x-1)/(x+1), 1/x$	12	3	(1, 1, 729)	196

Table 1: Explicit models. The full primitive matrix lists are in [Appendix A](#).

7 The cases of orders 2, 3, 4, and 6

The four smaller orders already illustrate the basic mechanisms without any difficult collision analysis. We record them explicitly before turning to the two dihedral cases in which the different residue-class branches interact nontrivially.

7.1 The involution and A365892

Let $G = C_2 = \langle \tau \rangle$ with $\tau(x) = x/(x-1)$. The orbit form factors as

$$F_n(X, Y) = (X - nY)((n-1)X - nY) = (n-1)X^2 - n^2XY + n^2Y^2. \quad (14)$$

It is primitive for every n , and therefore

$$H_{C_2}(n) = 2n^2 + |1 - n|. \quad (15)$$

The parameter $n = 1$ gives $-XY + Y^2$ and height 2; its orbit contains ∞ .

For $k \geq 1$, the two tails of the integer line give the quadratic sequences

$$H_{C_2}(-k) = 2k^2 + k + 1, \quad H_{C_2}(k+1) = 2k^2 + 5k + 2.$$

A direct comparison shows that these two sequences interlace. Hence, after omitting the exceptional value 2, the increasing spectrum begins

$$1, 4, 9, 11, 20, 22, 35, 37, \dots,$$

which is OEIS A365892 [\[7\]](#). In particular,

$$N_{C_2}^{\text{ht}}(X) = \sqrt{2} X^{1/2} + O(1).$$

A short independent proof of the equality $A365892(n) = A266257(n)$ for $n \geq 2$ is included in the computational supplement; see also OEIS A266257 [\[6\]](#).

7.2 A cyclic group of order 3

Let $G = C_3$ be generated by $T(x) = 1/(1-x)$. Direct multiplication gives

$$\begin{aligned} F_n(X, Y) = & -n(n-1)X^3 + (n^3 - 3n + 1)X^2Y \\ & + (-n^3 + 3n^2 - 1)XY^2 - n(n-1)Y^3. \end{aligned} \quad (16)$$

The form is primitive. Coefficient reversal under $n \mapsto 1 - n$ gives

$$H_{C_3}(n) = H_{C_3}(1 - n).$$

Writing $q = \max(n, 1 - n)$, a sign check yields

$$H_{C_3}(n) = \begin{cases} 2, & q = 1, \\ 10, & q = 2, \\ 2q^3 - q^2 - 5q + 2, & q \geq 3. \end{cases} \quad (17)$$

The final polynomial is strictly increasing for $q \geq 3$, so

$$N_{C_3}^{\text{ht}}(X) = 2^{-1/3}X^{1/3} + O(1). \quad (18)$$

Notice that the height symmetry $n \leftrightarrow 1 - n$ is not, in general, an equality of C_3 -orbits. It is an additional affine resonance and halves the parameter main constant.

7.3 The Klein four group

Let $G = D_2 = \langle x \mapsto -x, x \mapsto 1/x \rangle$. Then

$$F_n = (nX - Y)(nX + Y)(X - nY)(X + nY), \quad (19)$$

and expansion gives

$$F_n = n^2X^4 - (n^4 + 1)X^2Y^2 + n^2Y^4.$$

It is primitive and

$$H_{D_2}(n) = (n^2 + 1)^2. \quad (20)$$

After the symmetry $n \leftrightarrow -n$,

$$N_{D_2}^{\text{ht}}(X) = X^{1/4} + O(1). \quad (21)$$

7.4 A dihedral group of order 6

Let $G = D_3 = \langle T, S \rangle$, where $T(x) = 1/(1 - x)$ and $S(x) = 1/x$. Put

$$t = n(n - 1) \in \{0, 2, 6, 12, \dots\}.$$

The seven coefficients of the orbit form, from X^6 to Y^6 , are

$$t^2, -3t^2, -t^3 + 3t^2 - 3t - 1, 2t^3 - t^2 + 6t + 2, \quad (22)$$

followed by the reverse sequence. Their signs are $+, -, -, +, -, -, +$ for the admissible t , with zero coefficients interpreted harmlessly. Hence

$$H_{D_3}(n) = 4t^3 + t^2 + 12t + 4. \quad (23)$$

This is strictly increasing in $t \geq 0$ and invariant under $n \mapsto 1 - n$. Therefore

$$N_{D_3}^{\text{ht}}(X) = 2^{-1/3}X^{1/6} + O(1). \quad (24)$$

8 The order-8 dihedral group: Pell collisions

The order-8 case is the first one in which the periodic content creates genuinely different height branches. Here the period is 2, and the interaction between the even and odd branches turns out to be exactly a Pell equation.

Let

$$R(x) = \frac{x-1}{x+1}, \quad S(x) = \frac{1}{x},$$

so that R has order 4, $S^2 = 1$, and $SRS = R^{-1}$. The generated group is D_4 of order 8. Its orbit form factors as

$$\begin{aligned} F_n = & (nX - Y)(nX + Y)(X - nY)(X + nY) \\ & \cdot ((n-1)X - (n+1)Y)((n+1)X - (n-1)Y) \\ & \cdot ((n-1)X + (n+1)Y)((n+1)X + (n-1)Y). \end{aligned} \quad (25)$$

The first four factors are primitive. Each of the last four has content 2 exactly when n is odd. By [theorem 3.2](#),

$$\gamma_{D_4}(n) = \begin{cases} 1, & n \text{ even,} \\ 16, & n \text{ odd.} \end{cases} \quad (26)$$

The expansion has only even powers:

$$\begin{aligned} F_n = & n^2(n^2 - 1)^2(X^8 + Y^8) - (n^8 + 14n^4 + 1)(X^6Y^2 + X^2Y^6) \\ & + (2n^8 + 14n^6 + 14n^2 + 2)X^4Y^4. \end{aligned}$$

The displayed signs are fixed, and the uncut coefficient norm is

$$\|F_n\|_1 = 4(n^2 + 1)^4.$$

Consequently

$$H_{D_4}(n) = \begin{cases} 4(n^2 + 1)^4, & n \text{ even,} \\ (n^2 + 1)^4/4, & n \text{ odd.} \end{cases} \quad (27)$$

The height depends only on $|n|$, so we may take an even representative $q \geq 0$ or an odd representative $p \geq 1$. Each branch is strictly increasing on these representatives. Thus repetitions within a branch are impossible, and a repeated height can only come from one even and one odd parameter. Such a collision is equivalent to

$$\frac{(p^2 + 1)^4}{4} = 4(q^2 + 1)^4.$$

All quantities are positive, so taking fourth roots gives

$$p^2 - 2q^2 = 1. \quad (28)$$

Here p is automatically odd and q even. The solution $(p, q) = (1, 0)$ gives the initial duplicated height 4. All remaining nonnegative solutions are exactly

$$p_k + q_k\sqrt{2} = (3 + 2\sqrt{2})^k, \quad k \geq 1. \quad (29)$$

Theorem 8.1 (Pell secondary term). *For the order-8 model above,*

$$N_{D_4}^{\text{ht}}(X) = \frac{1 + \sqrt{2}}{2^{5/4}} X^{1/8} - \frac{\log X}{8 \log(3 + 2\sqrt{2})} + O(1). \quad (30)$$

Proof. The even branch contributes

$$\frac{1}{2^{5/4}} X^{1/8} + O(1),$$

and the positive odd branch contributes

$$2^{-3/4} X^{1/8} + O(1) = \frac{\sqrt{2}}{2^{5/4}} X^{1/8} + O(1).$$

Apart from the initial duplicate at height 4, every Pell solution (29) produces exactly one duplicated height, and there are no other duplicates after choosing the sign representatives. Since $p_k \asymp (3 + 2\sqrt{2})^k$, the corresponding height is $\asymp (3 + 2\sqrt{2})^{8k}$. Thus the number of duplicated heights at most X is

$$\frac{\log X}{8 \log(3 + 2\sqrt{2})} + O(1).$$

Subtracting these duplicates proves the result. $\square$

9 The order-12 dihedral group

The order-12 example has a more complicated content function but, in contrast with the Pell case, only finitely many cross-branch collisions. The calculation is longer, so we first reduce the coefficient norm to a single polynomial in $t = n(n - 1)$.

Let

$$R(x) = \frac{2x - 1}{x + 1}, \quad S(x) = \frac{1}{x}.$$

Then R has order 6, $S^2 = 1$, and $SRS = R^{-1}$, so $G = \langle R, S \rangle \cong D_6$ has order 12.

Among the twelve primitive matrix representatives in Appendix A, six have determinant 1 and six have determinant 3. For each determinant-3 factor, the gcd in (2) equals 3 precisely when $n \equiv 2 \pmod{3}$. Therefore

$$\gamma_{D_6}(n) = \begin{cases} 729, & n \equiv 2 \pmod{3}, \\ 1, & n \not\equiv 2 \pmod{3}. \end{cases} \quad (31)$$

Put

$$t = n(n - 1), \quad U = X^2 + Y^2, \quad V = XY.$$

The uncut orbit form admits the compact factorization

$$F_n = Q_1 Q_2 Q_3, \quad (32)$$

where

$$\begin{aligned} Q_1 &= tU^2 + (1-t)UV - (t^2 + 2t + 2)V^2, \\ Q_2 &= (t-2)(4t+1)U^2 - (4t+1)(5t-1)UV \\ &\quad + (25t^2 + 26t + 10)V^2, \\ Q_3 &= t(t-2)U^2 + 2(4t+1)UV - (2t+1)(2t+5)V^2. \end{aligned}$$

For the admissible values $t = 0, 2, 6, 12, \dots$, the coefficient of $X^{12-j}Y^j$ is nonnegative for $j \equiv 0 \pmod{3}$ and nonpositive otherwise. Appendix B gives a finite polynomial certificate.

Let ω be a primitive cube root of unity. A root-of-unity filter gives

$$\|F_n\|_1 = -\frac{1}{3}F_n(1, 1) + \frac{2}{3}(F_n(1, \omega) + F_n(1, \omega^2)). \quad (33)$$

Using (32),

$$\begin{aligned} F_n(1, 1) &= t^2(t-2)^2(4t+1), \\ F_n(1, \omega) &= F_n(1, \omega^2) = (t+7)(3t+1)(t^2+3)(49t^2+20t+7). \end{aligned}$$

Hence

$$\|F_n\|_1 = P(t), \quad (34)$$

where

$$\begin{aligned} P(t) &= 196t^6 + 1516t^5 + 1665t^4 + 4940t^3 \\ &\quad + 3280t^2 + 1176t + 196. \end{aligned} \quad (35)$$

Combining (31) and (34) gives

$$H_{D_6}(n) = \begin{cases} P(n(n-1)), & n \not\equiv 2 \pmod{3}, \\ P(n(n-1))/729, & n \equiv 2 \pmod{3}. \end{cases} \quad (36)$$

The symmetry $n \mapsto 1-n$ preserves both t and the residue condition. Choose $n \geq 1$. The three residue classes give the strictly increasing branches

$$\begin{aligned} A(k) &= P(9k^2 - 3k), & k &\geq 1, \\ B(k) &= P(9k^2 + 3k), & k &\geq 0, \\ C(k) &= \frac{P(9k^2 + 9k + 2)}{729}, & k &\geq 0. \end{aligned} \quad (37)$$

Since P has positive coefficients, it is strictly increasing on $[0, \infty)$. Thus $A(k) = B(\ell)$ would imply

$$9k^2 - 3k = 9\ell^2 + 3\ell,$$

or $3(k+\ell)(3k-3\ell-1) = 0$, which has no solutions in the indicated ranges.

There are no collisions between A and B , so the only remaining question is whether a value on the content-729 branch C can also occur on one of the

uncut branches. Since P is increasing, both possibilities reduce to the same integer equation

$$P(v) = 729P(u), \quad u, v \in \mathbb{Z}_{\geq 0}. \quad (38)$$

The following identities are exact:

$$729P(u) - P(3u + 1) = 3(150336u^5 + 75600u^4 + 924300u^3 + 658470u^2 + 247796u + 43305) > 0, \quad (39)$$

$$P(3u + 2) - 729P(u) = -18360u^2(u - 6)(9u^2 - 6u - 2), \quad (40)$$

$$P(3u + 3) - 729P(u) = 120528u^5 + 2906280u^4 + 3613140u^3 + 4674870u^2 + 2844108u + 669877 > 0. \quad (41)$$

For every $u \geq 1$, the factor $9u^2 - 6u - 2$ is positive. Hence for $1 \leq u \leq 5$ the right-hand side of (40) is positive, while (39) is always positive; therefore

$$P(3u + 1) < 729P(u) < P(3u + 2).$$

For $u \geq 7$, the right-hand side of (40) is negative, and (41) is positive; therefore

$$P(3u + 2) < 729P(u) < P(3u + 3).$$

Because P is strictly increasing, neither interval contains another value $P(v)$ with integer v . At $u = 0$ and $u = 6$, (40) vanishes and gives $v = 2$ and $v = 20$, respectively.

Theorem 9.1 (Exact D_6 collision classification). *The nonnegative integer solutions of (38) are exactly*

$$(u, v) = (0, 2), \quad (u, v) = (6, 20).$$

Consequently the only heights repeated across distinct residue-class branches are

$$196 \quad \text{and} \quad 24283204.$$

Proof. The two strict bracketing inequalities exclude every $u \notin \{0, 6\}$. For $u = 0$ and $u = 6$, the equality in (40) gives $v = 3u + 2$, hence $v = 2$ and $v = 20$. Direct substitution yields $B(0) = C(0) = 196$ and $A(1) = C(1) = 24283204$. $\square$

Corollary 9.2 (Exact count and asymptotic). *For every X ,*

$$N_{D_6}^{\text{ht}}(X) = \#\{k \geq 1 : A(k) \leq X\} + \#\{k \geq 0 : B(k) \leq X\} + \#\{k \geq 0 : C(k) \leq X\} - \mathbf{1}_{X \geq 196} - \mathbf{1}_{X \geq 24283204}. \quad (42)$$

In particular,

$$N_{D_6}^{\text{ht}}(X) = \frac{\frac{2}{3} + \frac{1}{\sqrt{3}}}{14^{1/6}} X^{1/12} + O(1). \quad (43)$$

Proof. The exact formula follows from [theorem 9.1](#). The leading coefficients of A and B are $196 \cdot 9^6$, whereas that of C is $196 \cdot 9^6 / 729$. Applying [theorem 4.1](#) gives

$$2(196 \cdot 9^6)^{-1/12} + (196 \cdot 9^6 / 729)^{-1/12} = \frac{2}{3 \cdot 14^{1/6}} + \frac{1}{\sqrt{3} \cdot 14^{1/6}},$$

which is the constant in (43). $\square$

10 Computational checks

The proofs above are independent of computer algebra. Exact symbolic computation was nevertheless useful for checking the longer explicit identities in Sections 7–9. Two separate implementations, one in Maple and one in Python/SymPy, were used to verify the primitive matrix representatives, the values of q_G and K_G , the explicit coefficient and height formulas, the D_4 factorization leading to the Pell equation, and the D_6 factorization and polynomial identities (39)–(41).

The accompanying supplementary file (Online Resource 1) contains the exact-arithmetic source code and the corresponding outputs. In particular, the sign pattern used in the root-of-unity evaluation of the D_6 coefficient norm is also checked directly there. These computations are included to make the explicit algebra reproducible; none of the general counting arguments in Sections 3–5 relies on numerical experimentation.

11 Further questions

Several natural variants are left open. The height considered here depends on the chosen rational coordinate. One could instead minimize it over an integral or rational conjugacy class and compare the resulting invariant with reduction theories for binary forms. It would also be interesting to replace the ℓ^1 norm by the maximum norm, the Euclidean norm, Mahler measure, or a projective Weil height, and to ask which parts of the periodic-content argument survive.

A second direction is to work over a number field. Coefficient content would then have to be formulated ideal-theoretically, while the parameter count would involve geometry of numbers at the archimedean places. Even the form of the leading constant is no longer as immediate as it is over $\mathbb{Q}$.

Finally, the examples of orders 8 and 12 suggest a Diophantine classification problem in their own right. For one rational dihedral model the branch intersections form a Pell family and contribute a logarithmic secondary term; for another model there are only two exceptional intersections. It would be useful to understand which collision patterns can occur for different rational embeddings of the same abstract finite group.

A Primitive matrix representatives

A tuple (a, b, c, d) denotes the primitive matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and the map $x \mapsto (ax + b)/(cx + d)$. The representatives used in the computations are:

$$\begin{aligned}
C_2 : & (1, 0, 0, 1), (1, 0, 1, -1). \\
C_3 : & (0, 1, -1, 1), (1, -1, 1, 0), (1, 0, 0, 1). \\
D_2 : & (0, 1, -1, 0), (0, 1, 1, 0), (1, 0, 0, -1), (1, 0, 0, 1). \\
D_3 : & (0, 1, -1, 1), (0, 1, 1, 0), (1, -1, 0, -1), (1, -1, 1, 0), \\
& (1, 0, 0, 1), (1, 0, 1, -1). \\
D_4 : & (0, 1, -1, 0), (0, 1, 1, 0), (1, -1, -1, -1), (1, -1, 1, 1), \\
& (1, 0, 0, -1), (1, 0, 0, 1), (1, 1, -1, 1), (1, 1, 1, -1). \\
D_6 : & (0, 1, -1, 1), (0, 1, 1, 0), (1, -2, -1, -1), (1, -2, 2, -1), \\
& (1, -1, 0, -1), (1, -1, 1, 0), (1, 0, 0, 1), (1, 0, 1, -1), \\
& (1, 1, -1, 2), (1, 1, 2, -1), (2, -1, 1, -2), (2, -1, 1, 1).
\end{aligned}$$

B A sign certificate for the D_6 form

Write

$$F_n(X, Y) = \sum_{j=0}^{12} c_j(t) X^{12-j} Y^j, \quad t = n(n-1).$$

The form is reciprocal, so $c_j = c_{12-j}$. The first seven coefficient polynomials are

$$\begin{aligned}
c_0 &= t^2(t-2)^2(4t+1), \\
c_1 &= -6t^2(t-2)^2(4t+1), \\
c_2 &= -4t^6 + 60t^5 - 375t^4 + 172t^3 + 24t^2 - 24t - 4, \\
c_3 &= 10(2t^6 - 8t^5 + 105t^4 - 20t^3 + 10t^2 + 12t + 2), \\
c_4 &= -5(5t^6 - 42t^5 + 345t^4 - 116t^3 + 3t^2 + 30t + 5), \\
c_5 &= -2(10t^6 + 312t^5 - 795t^4 + 956t^3 + 402t^2 + 60t + 10), \\
c_6 &= 58t^6 + 912t^5 - 1245t^4 + 2852t^3 + 1434t^2 + 348t + 58.
\end{aligned}$$

At the two exceptional admissible values $t = 0$ and $t = 2$ the signs are immediate by substitution. Every remaining admissible value has $t \geq 6$. Write $t = u + 6$ with $u \geq 0$. After multiplying by the predicted sign, each coefficient polynomial becomes a polynomial in u with nonnegative coefficients. For example,

$$\begin{aligned}
-c_2(u+6) &= 4u^6 + 84u^5 + 735u^4 + 4508u^3 \\
&\quad + 26040u^2 + 102984u + 168196, \\
c_3(u+6) &= 20u^6 + 640u^5 + 9450u^4 + 82600u^3 \\
&\quad + 439300u^2 + 1301640u + 1632980, \\
-c_5(u+6) &= 20u^6 + 1344u^5 + 27930u^4 + 274792u^3 \\
&\quad + 1428420u^2 + 3819144u + 4167380, \\
c_6(u+6) &= 58u^6 + 3000u^5 + 57435u^4 + 551852u^3 \\
&\quad + 2881290u^2 + 7865700u + 8854042.
\end{aligned}$$

The omitted cases are either shorter or follow by the reciprocity $c_j = c_{12-j}$; their explicit expansions are included in the supplement. Hence the coefficient of $X^{12-j}Y^j$ is nonnegative exactly for $j \equiv 0 \pmod{3}$ (with the possible zero coefficients at the exceptional values already accounted for).

Statements and Declarations

Funding. No funds, grants, or other support was received.

Competing interests. The author has no relevant financial or non-financial interests to disclose.

Author contributions. The author is solely responsible for the conception of the study, mathematical analysis, computational verification, and preparation of the manuscript.

Code and data availability. Exact-arithmetic Maple and Python/SymPy source files supporting the explicit calculations in Section 10 are supplied as supplementary material (Online Resource 1). They verify the displayed algebraic identities and the finite sign checks used in the examples. No external research dataset was used.

Use of AI-assisted tools. During the development and preparation of this work, the author used OpenAI's ChatGPT extensively for mathematical exploration, the development and checking of proof arguments, symbolic computation and Maple/Python/SymPy code, literature organization, and drafting and language editing. The author determined the research direction, reviewed the mathematical statements and proofs, executed or reviewed the computational checks, and reviewed the submitted text. The author takes full responsibility for the content of the article.

References

- [1] A. Beauville, *Finite subgroups of $\mathrm{PGL}_2(K)$* , in *Vector Bundles and Complex Geometry*, Contemp. Math. 522, Amer. Math. Soc., 2010, 23–29. <https://doi.org/10.1090/comm/522/10289>.
- [2] Y. F. Bilu and R. F. Tichy, *The Diophantine equation $f(x) = g(y)$* , Acta Arith. 95 (2000), no. 3, 261–288. <https://doi.org/10.4064/aa-95-3-261-288>.
- [3] E. Bombieri and J. Pila, *The number of integral points on arcs and ovals*, Duke Math. J. 59 (1989), no. 2, 337–357. <https://doi.org/10.1215/S0012-7094-89-05915-2>.

- [4] R. Gow and G. McGuire, *Invariant rational functions, linear fractional transformations and irreducible polynomials over finite fields*, Finite Fields Appl. 79 (2022), Article 101991. <https://doi.org/10.1016/j.ffa.2021.101991>.
- [5] B. Hutz and M. Stoll, *Smallest representatives of $\mathrm{SL}(2, \mathbb{Z})$ -orbits of binary forms and endomorphisms of $\mathbb{P}^1$* , Acta Arith. 189 (2019), no. 3, 283–308. <https://doi.org/10.4064/aa180618-9-12>.
- [6] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A266257, <https://oeis.org/A266257>.
- [7] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A365892, <https://oeis.org/A365892>.
- [8] M. Schulz, *Rational transformations and invariant polynomials*, arXiv:2306.13502, 2023.
- [9] J.-W. M. van Ittersum, *A group-invariant version of Lehmer’s conjecture on heights*, J. Number Theory 171 (2017), 145–154. <https://doi.org/10.1016/j.jnt.2016.07.015>.